

Cornell University researchers discover 'code-poisoning' attack

Cornell Tech researchers have claimed to have uncovered a new type of online attack that can manipulate natural-language modelling systems and evade any known defence. The team forecasts possible consequences ranging from modifying movie reviews to manipulating investment banks' machine learning models so that they ignore negative news coverage that would affect a specific company's stock.

Without any access to the original code or model, these backdoor attacks can upload malicious code to open source sites frequently used by many companies and programmers.

In a new paper, researchers found the implications of these types of hacks – which they call 'code poisoning' – to be wide-reaching for everything from algorithmic trading to fake news and propaganda.

"With many companies and programmers using models and codes from open source sites on the Internet, this research shows how important it is to review and verify these materials before integrating them into your current system," said Eugene Bagdasaryan, a doctoral candidate at Cornell Tech.

"If hackers are able to implement code poisoning," Bagdasaryan said, "they could manipulate models that automate supply chains and propaganda, as well as resume-screening and toxic comment deletion."

As opposed to adversarial attacks, which require knowledge of the code and model to make modifications, backdoor attacks allow the hacker to have a large impact without actually having to directly modify the code and models.

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"So far, we have seen two fairly compartmentalised efforts to evolve the use of bare metal with all of its benefits and to modernise Kubernetes management. The problem has been that each of these initiatives has been siloed thus far. With the Spectro Cloud Cluster API provider for MaaS, we have built the bridge between the two—this will enable the whole market to expand," he added.

This latest open source contribution from Spectro Cloud bridges two paradigm shifts for Kubernetes: Cluster API, which enables life cycle management in a declarative manner for Kubernetes environments, and bare metal performance. It allows any team to use CNCF's Cluster API to provision Kubernetes clusters on bare metal servers in any data centre, and manage them just like any other Kubernetes deployment target, without the extra hassle and risks, or additional technical skillsets.

The open source software, under the Apache licence and available for anyone to download, can be used by IT and development teams that are interested in managing Kubernetes environments at scale and in extending to bare metal servers.

"Kubernetes can be challenging, let alone deploying directly on bare metal servers. What we have built is essentially a way for any team to deploy and manage bare metal Kubernetes just like any other cluster. This open source contribution, along with the Spectro Cloud platform, further demonstrates our work in bringing life to the CNCF Cluster API project," added Saad Malik, CTO, Spectro Cloud.

Debian 11 'bullseye' Linux distro launched with new features

The project behind the Debian Linux distribution has released its stable version Debian 11, code named 'bullseye'. This version was two years, one month and nine

days in the making, and is said to be supported for the next five years.

A Debian release is usually considered a big deal, with support coming from its user community rather than a company. Debian is also the core operating

system for other Linux distributions such as Canonical's Ubuntu and Raspberry Pi OS.

The Debian 11 release comes with several desktop environments, including GNOME 3.38, KDE Plasma 5.20, LXDE 11, LXQt 0.16, MATE 1.24, and Xfce 4.16.

This release contains over 11,294 new packages for a total count of 59,551 packages, along with a significant reduction of over 9,519 packages which were marked as 'obsolete' and removed. 42,821 packages were updated and 5,434 packages remained unchanged.

Version 11 'bullseye' is the first Debian release to provide a Linux kernel with support for the exFAT (extended file allocation table) file system, and defaults to using it to mount exFAT file systems. "Consequently, it is no longer required to use the file system-in-user space implementation provided via the exfat-fuse package. Tools for creating and checking an exFAT file system are provided in the exfatprogs package," the Debian team said.

Debian 11 introduces a new package, ipp-usb, which uses the vendor-neutral Internet Printing Protocol-over-USB (IPP-over-USB) protocol supported by many modern printers.

"This allows a USB device to be treated as a network device. The official SANE driverless backend is provided by sane-escl in libsane1, which uses the eSCL protocol," explain the Debian team.

